

Indoor Air Quality in the age of Covid-19

What you should know when considering measures to combat the virus.

There is a lot of talk about what non-healthcare facilities can do to prepare for operation during a pandemic and a lot of measures circling around indoor air quality and health in buildings. To help you weed through this I have compiled a list of measures frequently discussed, what they can accomplish and what you need to consider as well as putting it in context of what is known about the virus. This article will not cover any new technology but rather known strategies and technologies that building owners can implement without major changes to systems. There is no silver bullet that will guaranty no infections. It is about creating a swiss cheese layer of measures to minimize risks of infection.

Increasing outside air

There is still a lot unknown about the disease Covid-19 and the virus SARS-CoV-2. Researchers and experts like Christian Drosten, virologist and corona virus expert, frequently remind us to go back to basics, when detail for this virus aren't yet known. For any virus, the following holds true: more virus concentration in the air or on surfaces the more likely it is that someone will get infected by breathing in the air/droplets or touching the surface. However, the level of dilution or concentration of the virus to be infectious is not yet known. There are also some studies shown that a lower dosage of the virus leads to milder infections.

SARS-CoV-2 is a single-stranded enveloped RNA virus¹ and belongs to the family of corona viruses. The virus becomes inactive after it is dried out. What is still up to debate is how long it takes for the virus to dry out under normal everyday circumstances. While there have been studies to look at this, they were mostly under laboratory conditions with a set temperature, humidity and air speed, as well as not considering sun light. Sun light and its UV lighting will not kill the virus, however exposure to sun light will dry out the virus quicker.

To dilute any possible air pollutant such as a virus aerosol a building can introduce more outside air, which can be done as follows:

1. Increase outside air setting and open the outside air dampers more
2. Use natural ventilation: open windows
3. Use fans to provide more fresh air and better air mixing

The higher the percentage of fresh outside air, which you bring in the building the less return air from spaces and the more dilution of air contaminants takes place. Most commonly outside air intake is set to meet code minimum and optimal efficiency. When you increase outside air intake your energy cost typically increases, because you are heating or cooling more unconditioned air. However, during the changeover of the season and during the night in summer, there is opportunity to bring in more fresh air without a large increase in energy costs. Otherwise in summer you can also increase the room temperature to offset the increase energy consumption. Doing that however should come with effective communication as to why you are doing it and you may also provide individual small desk fans, which

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https://www.hopkinsguides.com/hopkins/view/Johns_Hopkins_ABX_Guide/540747/all/Coronavirus_COVID_19_SARS_CoV_2

also increases dilution of contaminants. Ideally place them so they point down, moving any droplets downwards to the floor and away from other people.

Commercial kitchens with exhaust hoods and make up air units (MAU) typically can increase outside air intake easily by turning on the kitchen hood. MAU are 100% outside air units, which bring in fresh air as air is exhausted by the kitchen hoods. Running the hoods on low setting or intermittently while they are not otherwise required to run increases fresh air into the space.

For simpler HVAC systems, ensure the system is set to have fans running continuously (ON) during occupied hours and not just cycle with occupancy (AUTO). Some units are controlled by a thermostat only and are often set to “Fan Auto”, which means they are cycling with occupancy only. Regardless of the amount of outside air more circulation by the AC fans, ceiling fans or else is diluting contaminant concentration in the space. This circulation without fresh outside air however achieve only a better distribution of the air contaminant through the building.

Research from Germany, the Technical University Berlin, has looked at aerosols in office spaces with and without AC. ² Simulations show that ventilated spaces fair better, diluting and removing aerosols far quicker.

Other means of getting more fresh air into the building is making use of natural ventilation, open the windows, or window fans bringing in fresh air and exhausting air. If using those strategies however consider placement. Do not place window fans such that air contaminant are being brought into the space. So, don't place window fans, where people outside sit and wait or close to exhaust air openings (toilet exhaust, kitchen exhaust). ASHRAE recommends and most codes require 8ft distance between exhaust and supply air intakes. Also keep in mind for air to naturally move from the inside to the outside the temperature difference is important. At the same indoor and outdoor temperature air movement is minimal and fans need to help move the air around.

A way to measure where you are and if your measures are doing well to achieve dilution of air contaminates, is to measure the CO₂. The closer you get with indoor CO₂ levels to outdoor CO₂ levels the better you are doing. 1000 ppm maximum is considered good indoor air quality, while you general want this to be between 600 to 800 ppm. Outdoor air typically is between 300 to 400 ppm CO₂. The advantage of using CO₂ is that it's also a great indicator of the occupancy density in a space. People breath out CO₂ increasing its concentration in the space if enough fresh air is delivered to the space the CO₂ level should not rise above 1000 ppm. Therefore, CO₂ levels will be able to help determine if you bring in enough outside air and also if you have too many people in a space and not enough fresh air. The American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHREA) is recommending the increase of outdoor air and the TU Berlin has look into the relation of CO₂ to aerosol concentration and as aerosols can't be easily measured in the air they found it to be a great indicator for the presence of aerosols. ³

² The German code for outdoor air ventilation is typically requiring more outdoor air than in the US. The same principles however apply. <https://www.youtube.com/watch?v=jGyTVJ05qC0>
<https://www.youtube.com/watch?v=lnj0WmhldZA&fbclid=IwAROLLjx2cyfglP63Eifbkln2QF9qOHRspnxoRER6rZJ574n84mf8xT4LajY>

³ <https://www.tu.berlin/en/research/themenportal-forschen/2020/august/faq-about-aerosols-as-regards-sars-cov-2/?fbclid=IwAR0-R5KFmJqNpnEMLkBWgchPM4QiewBgtZOLSfCkwtws4gII0Evl3ur61s>

You can use handheld devices or wall mounted sensors to monitor the CO2 level and provide an alarm when levels rise above 1000 ppm. Of course, installing such sensors, which automatically adjust air supply is an option as well, however often costly in a retrofit situation. VOC sensors can help as well. They are measuring other air contaminants, which are typically emitted from building materials, printers or other equipment and human activity in the space. However, CO2 is a far better indicator for fresh air and occupancy. When measuring CO2 ensure you follow recommended height (typically 3-6ft) and manufacturer instruction for correct measuring, sampling strategies or enlist an indoor air quality professional to help you do it correctly and install sensors in the right spot.

Indoor temperature and humidity

There is some research with regards to temperature and humidity and its effect on the virus. For many experts it is still too little to provide a definitive answer for optimal operating temperature or humidity in indoor spaces. While heating up air is more likely to dry-out droplets and aerosols making the virus inactive, dry indoor air however also makes humans more susceptible for infections drying out their nasal membranes. What might be bad for the virus, might also be bad for humans comfort and health with regards to temperature and humidity in indoor spaces. So keep temperatures at a comfortable level 70-78F is best and humidity between 40-60%.

Air filtration

With regards to air filtration one must understand a few details. While the virus is only 0.125 microns large, it travels in droplets (1-5mm) and it may also travel in aerosols as small as 0.3 micron. The larger the particle and in this case droplet the quicker it sinks to the surface in still air. The smaller droplets turn into aerosols, which remain in the air for a significant longer time. The size of aerosols of concern is noted as 1.0-5.0 microns⁴. At the same time though these particles keep shrinking and eventually drying out.

HVAC filtration in central air system is rated from MERV 1-19. MERV 13 filter or higher can filter 90% or more particle of the size of 1.0-3.0 micron. A HEPA filter starts at MERV 17 with a filter effectiveness of 99% for particles size 0.3-1.0, 1.0-3.0 and 3-10 microns. Replacing your filter with anything below MERV 10, which is 50-64% effective for particles size 1.0-3.0 microns, is not going to remove the virus effectively enough to make a difference. ASHRAE recommends the use of MERV 13 or higher in non-healthcare buildings⁵. However, when upgrading the filter efficiency of your system you may run into several issue:

- Size of the filter spot: higher rated filters are generally thicker and may not fit into existing equipment without changes
- Fan sizing: due to the increase in resistance with a higher rated filter the fans will need to work harder to push air through them. This results in increased wear and tear on the fans, may even break them and/or reduce air flow in the system
- Energy efficiency: the increase in resistance may result in higher energy costs

Maintaining the filter and changing them frequently maybe even more often than recommended, needs to take place for this measure to be effective and also not unnecessarily increase energy consumption.

⁴ <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7094991/>

⁵ <https://www.ashrae.org/technical-resources/resources>

Keep in mind that in an environment that produces high amounts of particles (cutting, mixing of dry ingredience, ect) those filters will need replaced even more often.

Standalone solutions like air purifier and air scrubbers can help as well. Look for equipment with HEPA filter. Air scrubber should be ozone free and if available use UVGI technology as well. Regular maintenance and filter replacement are important as well or otherwise renders the equipment ineffective.

When selecting an air purifier look for the following:

1. True HEPA filter: MERV 17 or higher
2. UV or other plasmawave technology needs to be CARB certified, which means save levels of ozone or none
3. Noise level, the quieter the better
4. Energy Star for energy efficiency, note equipment that's EnergyStar certified also meets requirements for save levels of ozone.
5. The size of the air purifier should be based on room size. However even an undersized air filter still helps.
6. Optional is an air quality monitor, which automatically adjust the air purifier. However, those sensors adjust speed based on VOCs and particle matters not CO2. If used in a classroom automatic adjustment should be turn off during occupied hours.
7. Also check price and availability of replacement filters.

UVGI – Ultraviolet germicidal irradiation

UVGI – Ultraviolet germicidal irradiation was been show effective in the past to destroy the tuberculosis virus. The UV light destroys the DNA / RNA of a bacteria or virus and renders it inactive. ASHRAE in its position paper on infectious aerosols⁶ notes UVGI as a measure, which “can kill or inactivate microorganisms”. The CDC notes “UVGI can be used as an adjunct air-cleaning measure, but it cannot replace HEPA filtration” (Guidelines for Environmental Infection Control in Health-Care Facilities 2003)⁷. It is not yet confirmed that UVGI will work effectively on CoVid-19. The basic premise of the technology and effectiveness on other virus would make it likely to work, however there are no studies yet.

UVGI or air scrubbers with UVGI requires professional installation and may or may not be possible for your HVAC system.

Case example – small restaurant

Note each restaurant's HVAC system and setup is different. Always review your current setup and HVAC system settings first and have an HVAC professional help you, where necessary. This example assumes a commercial kitchen and early summer to summer operation in Ohio climate.

⁶ <https://www.ashrae.org/technical-resources/resources>

⁷ <https://www.cdc.gov/niosh/docs/2009-105/pdfs/2009-105.pdf>

Risk mitigation measures:

Kitchen:

- Set kitchen hoods on low or run intermittently while the kitchen is occupied and prior to start of business each day. → brings in more fresh air
- Make-up air units (MAU) bring in 100% outside air. Replacing their filter improves Indoor Air Quality, however it does not mitigate air contaminants inside the space.

Seating /Reception:

- Ensure your HVAC system is working properly and as designed. Have your system retro-commissioned if you have doubts and your system is older than 6 years or your space has undergone changes (occupancy increase, moved walls, changed use)
- If possible, increase outside air setpoint for the system and check economizer settings. Have an HVAC professional help you with that.
- Replace return air filter with MERV 13 or higher, if possible
- Run your AC before beginning of business to flush out the space or open windows to provide more outside air.
- Turn on ceiling fans if you have them
- Install window fans to exhaust and bring in more air or make use of cross ventilation by opening window on opposite ends of the space.
- Measure CO2 in the space as one-time measure to determine, if you are having outside air issues and/or install continuous measurement devices with alarms or notifications

Summary

Using a combination of those measures in addition to social distancing measures as required or recommended by your state will not eliminate the risk of transmitting the virus. They will however reduce the risk, if done as recommended.

If you want to learn more ASHRAE has published a resource page for Covid

<https://www.ashrae.org/technical-resources/resources>

A very in depth resource for aerosols indoors can be found here

<https://www.tu.berlin/en/research/themenportal-forschen/2020/august/faq-about-aerosols-as-regards-sars-cov-2/?fbclid=IwAR0-R5KFmJqNpnEMlkBWgchPM4QiewBgtZOLSfCkwtws4gII0Evl3ur61s>

About the Author

Susann Geithner is an international green building consultant, LEED AP and WELL AP consulting projects and buildings with regards to improving their sustainability. Indoor Air Quality consulting and measurements, optimization of HVAC systems and advising architects on product and material selection to improve sustainability and decrease indoor air pollutants is part of her practice, which she has been doing since 2007. Some of her largest projects include Orlando International Airport Terminal C, The Circle at Zurich Airport, Big River Steel in Arkansas, Maurer Center at Bowling Green University, Tri-C STEM building in Parma and the Squire at Frankfurt Airport. She holds a degree in architecture from the University in Applied Science in Dessau, Germany and a Master from the Bauhaus University in Weimar, Germany. As an expert in her field she has taught at the Virginia

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